

ARBORI BINARI

1. METODE DE PARCURGERE

1.1 Parcurgerea în preordine

% (rădăcina, subarborele din stânga, subarborele din dreapta)

preordine(null).

preordine(tree(S,R,D)):- write(' '), write(R), preordine(S), preordine(D).

1.2 Parcurgerea în inordine

% (subarborele din stânga, rădăcina, subarborele din dreapta)

inordine(null).

inordine(tree(S,R,D)):- inordine(S), write(' '), write(R), inordine(D).

1.3 Parcurgerea în postordine

% (subarborele din stânga, subarborele din dreapta, rădăcina)

postordine(null).

postordine(tree(S,R,D)):- postordine(S), postordine(D), write(' '), write(R).

1.4 Suma etichetelor numerice asociate nodurilor din arborele binar

suma(null, 0).

suma(tree(R,null,null), R).

suma(tree(S,R,D), SumTree):- suma(S, STS), suma(D, STD), SumTree is STS+STD+R.

sum_nodeTree(A,R):- arbore(A), suma(A,R).

arbore(tree(tree(null,3,tree(null,4,null)),5,tree(tree(tree(null,6,tree(null,7,null)),8,null))).

2. Arbori binari ordonati ($S < R < D$)

2.1 Adăugarea/inserarea unei frunze în arbore

addleaf(null,X, tree(null,X,null)).

addleaf(tree(S, X, D), X, tree(S, X, D)).

addleaf(tree(S, R, D), X, tree(S1, R, D)):- $X < R$, addleaf(S, X, S1).

addleaf(tree(S, R, D), X, tree(S, R, D1)):- $X > R$, addleaf(D, X, D1).

Executie: arbore(A), addleaf(A,2,R).

2.2 Ștergerea unei frunze în arbore

delleaf(Arb,X,ArbNew):- addleaf(ArbNew,X,Arb).

2.3 Ștergerea unui nod intern din arbore

del(tree(null, X, R), X, R).

del(tree(L, X, null), X, L).

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del(tree(Left, X, Right), X, tree(Left, Y, Right1)):- delmin(Right, Y, Right1).
del(tree(Left, Root, Right), X, tree(Left1, Root, Right)) :- Root>X, del(Left, X,
Left1).
del(tree(Left, Root, Right), X, tree( Left, Root, Right1) ) :- X>Root, del(Right, X,
Right1).
delmin(tree(null, Y, R), Y, R).
delmin(tree(Left, Root, Right), Y, t(Left1, Root, Right)):- delmin(Left, Y, Left1).
Executie: arbore(A), del(A,5,R).

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2.4 Adăugarea unui nod intern din arbore

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add(T, X, T1):- addroot(T, X,T1).
add(tree(Left, Y, Right), X, tree(Left1, Y, Right)) :- Y>X, add(Left, X, Left1).
add(tree(Left, Y, Right), X, tree( Left, Root, Right1) ) :- X>Y, add(Right, X,
Right1).

addroot(null, X, tree(null, X, null)).
addroot(tree(L, X, R), X, tree(L, X, R)).
addroot(tree(Left, Y, Right), X, tree(Left1, X, tree(Left2, Y, Right))):- Y>X,
addroot(Left, X, tree(Left1, X, Left2)).
addroot(tree(Left, Y, Right), X, tree(tree(Left, Y, Right1),X,Right2)):- Y<X,
addroot(Right, X, tree(Right1, X, Right2)).

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Executie: arbore(A), add(A,5.5,R).